

SAT 9
Math Module 3

Question # ID
SAT9 M 3.1 35bec412

73, 74, 75, 77, 79, 82, 84, 85, 91
What is the median of the data shown?

SAT9 M 3.2 4ff7b652

Right triangles LMN and PQR are similar, where L and M correspond to P and Q , respectively. Angle M has a measure of 53° . What is the measure of angle Q ?

- A. 37°
- B. 53°
- C. 127°
- D. 143°

SAT9 M 3.3 a09ma303

x	y
1	11
2	19
3	a

The table shows three values of x and their corresponding values of y for the equation $y = 4(2)^x + 3$. In the table, a is a constant. What is the value of a ?

(A) 67

(C) 32

(B) 35

(D) 27

SAT9 M 3.4 5a7ab8e8

$$66x = 66x$$

How many solutions does the given equation have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

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Question # **ID**
SAT9 M 3.5 c048055c

A model predicts that the population of Springfield was 15,000 in 2005. The model also predicts that each year for the next 5 years, the population p increased by 4% of the previous year's population. Which equation best represents this model, where x is the number of years after 2005, for $x \leq 5$?

- A. $p = 0.96(15,000)^x$
- B. $p = 1.04(15,000)^x$
- C. $p = 15,000(0.96)^x$
- D. $p = 15,000(1.04)^x$

SAT9 M 3.6 548a4929

The function h is defined by $h(x) = 4x + 28$. The graph of $y = h(x)$ in the xy -plane has an x -intercept at $(a, 0)$ and a y -intercept at $(0, b)$, where a and b are constants. What is the value of $a + b$?

- A. 21
- B. 28
- C. 32
- D. 35

SAT9 M 3.7 52931bfa

Which expression is equivalent to $\frac{8x(x-7)-3(x-7)}{2x-14}$, where $x > 7$?

- A. $\frac{x-7}{5}$
- B. $\frac{8x-3}{2}$
- C. $\frac{8x^2-3x-14}{2x-14}$
- D. $\frac{8x^2-3x-77}{2x-14}$

SAT9 M 3.8 f329442c

Circle A has a radius of $3n$ and circle B has a radius of $129n$, where n is a positive constant. The area of circle B is how many times the area of circle A ?

- A. 43
- B. 86
- C. 129
- D. 1,849

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Question # ID
SAT9 M 3.9 24cec8d1

A circle has center O , and points R and S lie on the circle. In triangle ORS , the measure of $\angle ROS$ is 88° . What is the measure of $\angle RSO$, in degrees? (Disregard the degree symbol when entering your answer.)

SAT9 M 3.10 03503d49

A business owner plans to purchase the same model of chair for each of the 81 employees. The total budget to spend on these chairs is \$14,000, which includes a 7% sales tax. Which of the following is closest to the maximum possible price per chair, before sales tax, the business owner could pay based on this budget?

- A. \$148.15
- B. \$161.53
- C. \$172.84
- D. \$184.94

SAT9 M 3.11 f811d345

A right triangle has legs with lengths of 24 centimeters and 21 centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$, where d is an integer, what is the value of d ?

SAT9 M 3.12 a09ma312

The positive number a is 230% of the number b , and a is 60% of the number c . If c is $p\%$ of b , which of the following is closest to the value of p ?

(A) 138

(B) 217

(C) 283

(D) 383

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Question # **ID**
SAT9 M 3.13 9b5b23fc

For $x > 0$, the function f is defined as follows:
 $f(x)$ equals 201% of x
Which of the following could describe this function?

- A. Decreasing exponential
- B. Decreasing linear
- C. Increasing exponential
- D. Increasing linear

SAT9 M 3.14 3008cfc3

x	y
k	13
$k + 7$	-15

The table gives the coordinates of two points on a line in the xy -plane. The y -intercept of the line is $(k - 5, b)$, where k and b are constants. What is the value of b ?

SAT9 M 3.15 cc7ffe02

Keenan made 32 cups of vegetable broth. Keenan then filled x small jars and y large jars with all the vegetable broth he made. The equation $3x + 5y = 32$ represents this situation. Which is the best interpretation of $5y$ in this context?

- A. The number of large jars Keenan filled
- B. The number of small jars Keenan filled
- C. The total number of cups of vegetable broth in the large jars
- D. The total number of cups of vegetable broth in the small jars

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Question # ID
SAT9 M 3.16 a09ma316

$$3x = 36y - 45$$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

(A) $x = 4y$

(B) $\frac{1}{3}x = 4y$

(C) $x = 12y - 15$

(D) $\frac{1}{3}x = 12y - 15$

SAT9 M 3.17 fbb96bb1

$$x - 29 = (x - a)(x - 29)$$

Which of the following are solutions to the given equation, where a is a constant and $a > 30$?

I. a

II. $a + 1$

III. 29

A. I and II only

B. I and III only

C. II and III only

D. I, II, and III

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Question # ID
SAT9 M 3.18 a09ma318

One gallon of stain will cover **170** square feet of a surface. A yard has a total fence area of w square feet. Which equation represents the total amount of stain S , in gallons, needed to stain the fence in this yard twice?

(A) $S = \frac{w}{170}$

(B) $S = 170w$

(C) $S = 340w$

(D) $S = \frac{w}{85}$

SAT9 M 3.19 a54753ca

In the xy -plane, the graph of the equation $y = -x^2 + 9x - 100$ intersects the line $y = c$ at exactly one point. What is the value of c ?

A. $-\frac{481}{4}$

B. -100

C. $-\frac{319}{4}$

D. $-\frac{9}{2}$

SAT9 M 3.20 a09ma320

The quadratic function g models the depth, in meters, below the surface of the water of a seal t minutes after the seal entered the water during a dive. The function estimates that the seal reached its maximum depth of **302.4** meters **6** minutes after it entered the water and then reached the surface of the water **12** minutes after it entered the water. Based on the function, what was the estimated depth, to the nearest meter, of the seal **10** minutes after it entered the water?

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Question # ID
SAT9 M 3.21 a09ma321

The area of a rectangular region is increasing at a rate of 250 square feet per hour. Which of the following is closest to this rate in square meters per minute? (Use 1 meter = 3.28 feet.)

(A) 0.39

(B) 1.27

(C) 13.67

(D) 23.24

SAT9 M 3.22 a35c7164

$$\begin{aligned}5x + 7y &= 1 \\ ax + by &= 1\end{aligned}$$

In the given pair of equations, a and b are constants. The graph of this pair of equations in the xy -plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

A. $10x + 7y = 1$
 $ax - 2by = 1$

B. $10x + 7y = 1$
 $ax + 2by = 1$

C. $10x + 7y = 1$
 $2ax + by = 1$

D. $5x - 7y = 1$
 $ax + by = 1$